

MODULE 33 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Portfolio Assessment and Wave Planning

Domain 4 begins. You cannot migrate what nobody can describe.

DOMAIN 4 of 4 EXAM WEIGHT 20% MODULE 33 of 44

An honest warning, up front

- ◆ **AWS Migration Hub closed to new customers on 7 November 2025.** Existing customers continue; no new features.
- ◆ **AWS Application Discovery Service closed on the same date.** Same terms.
- ◆ **Both remain on the exam's in-scope service list**, so both remain correct exam answers.
- ◆ **AWS Transform** is the stated replacement, carrying discovery, dependency mapping, wave planning and strategy recommendations. It is **not** on the in-scope list.

ON THE EXAM

Simple AD and App Runner had this problem individually. Here it affects the two central services of an entire domain.

Nobody knows what they are running.
**Discovery is not a formality; it is the reason
migrations slip.**

Which tool answers which question

QUESTION	TOOL	NOTE
What servers do we have?	Application Discovery Service	Agent or agentless
What talks to what?	Discovery Agent	Connections and processes
What will it cost on AWS?	Migration Evaluator	TCO and what-if scenarios
How should each app move?	Strategy Recommendations	A path per application
How is it going?	Migration Hub	Aggregates. Moves nothing

ON THE EXAM

Migration Hub is a free dashboard that performs no migration. It shows MGN, DMS and partner status in one place.

Agent or agentless discovery?

AGENTLESS COLLECTOR

A VMware appliance

- One appliance, no software on servers
- VM inventory, profile and utilisation
- Database metadata and network traffic
- Cannot see inside the guest OS

DISCOVERY AGENT

Software on each server

- Network connections, per process
- Running processes and performance
- This is what gives dependency mapping
- Must be installed and permitted

THE TIEBREAKER

Dependency mapping needs the agent. If a scenario asks what talks to what, agentless collection alone will not answer it.

Building the business case

- ◆ **Migration Evaluator** produces a directional TCO comparison against the current on-premises cost.
- ◆ It models **what-if purchasing scenarios**, so the board sees On-Demand against commitments rather than one number.
- ◆ It analyses **Windows and SQL Server licensing** specifically, which is often the largest single line in the case.
- ◆ It can consume existing third-party inventory data, or Application Discovery Service data, instead of its own collector.

ON THE EXAM

The business case is a different artefact from the technical plan. A board approves money; an architect approves a design.

Planning the waves

- ◆ **A wave is a group of applications migrated together**, chosen because they depend on each other more than on anything outside.
- ◆ **Wave one should be low risk and low dependency**, because its real purpose is to prove the process and train the team.
- ◆ **Splitting a dependency across waves** means running traffic between on-premises and AWS for however long the gap lasts.
- ◆ Sequence by dependency first, then by business risk. Technical difficulty is a poor primary sort.

ON THE EXAM

The first wave's deliverable is a working process, not a migrated application. Choose something you can afford to get wrong.

Migration Hub mechanics

- ◆ **It aggregates, it does not migrate.** Status arrives from MGN, DMS and supported partner tools.
- ◆ **The home region is set once and cannot be changed.** All Migration Hub data lives there, whatever Region you migrate into.
- ◆ Group servers into applications, then track progress per application rather than per server.
- ◆ It is free. The cost is in the tools it tracks, not in the tracking.

ON THE EXAM

The home region catches people. Choosing it casually on day one is a decision you live with for the whole programme.

Reading the scenario

THE SCENARIO SAYS

We need to know what talks to what

No software may be installed on servers

The board needs a cost comparison

Track progress across several migration tools

Which applications should move together?

Recommend a migration path per application

IT IS POINTING AT

Discovery Agent, not agentless

Agentless Collector

Migration Evaluator

Migration Hub

Dependency-driven wave planning

Strategy Recommendations

ON THE EXAM

Assessment questions are usually about which question is being asked, not which tool is best. Match the tool to the question.

Meridian Freight is planning to migrate 1,200 on-premises servers. The existing CMDB is four years old and known to be inaccurate. Before planning waves, the team must establish which servers communicate with which, including undocumented dependencies. Security has approved installing software on the servers.

What should the solutions architect recommend?

- ☐ A Deploy the Application Discovery Agent to the servers, to collect network connections and running processes
- ☐ B Deploy the Agentless Collector into vCenter
- ☐ C Export the CMDB and validate it in a workshop with application owners
- ☐ D Run Migration Evaluator to produce a dependency map

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Discovery Agent

Only the agent sees network connections per process on each server, which is what dependency mapping requires. Agents are permitted here.
- B** Agentless Collector

Excellent for inventory and utilisation, and it cannot see inside the guest OS. Process-level connections are exactly what it misses.
- C** Validate the CMDB

Asks people about undocumented dependencies. Undocumented means nobody wrote them down, usually because nobody knew.
- D** Migration Evaluator

Builds a business case and a TCO model. It answers what it will cost, not what connects to what.

THE SIGNAL

Two details decide it: undocumented dependencies, and agents are permitted. The second exists to remove the usual objection.

A bank must assess 600 VMware virtual machines before migration. Its change policy forbids installing any new software on production servers, and the approval process for an exception would take nine months.

The team needs inventory, utilisation data and database metadata.

What should the solutions architect recommend?

- ☐ A Deploy the Application Discovery Service Agentless Collector as a vCenter appliance
- ☐ B Begin the exception process so agents can be installed
- ☐ C Deploy the legacy Discovery Connector
- ☐ D Estimate the inventory from purchase records and vendor documentation

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Agentless Collector in vCenter
One appliance, nothing on the servers, and it collects exactly what is asked for: inventory, utilisation and database metadata.
- B** Pursue an exception
Nine months before discovery starts, to obtain detail the scenario does not ask for. It does not need dependency mapping.
- C** Discovery Connector
The predecessor, and it reached end of support in November 2025. Application Discovery Service no longer accepts its data.
- D** Estimate from records
Purchase records show what was bought, not what runs or how heavily. Utilisation cannot be inferred from an invoice.

THE SIGNAL

Pair this with scenario one. Same choice, opposite answer, because agents are forbidden and no dependency mapping is required.

A board will not approve a migration programme without a comparison of current on-premises total cost against projected AWS cost, including Windows and SQL Server licensing options. The company already has two years of inventory and utilisation data from a third-party monitoring tool.

The CFO wants to see more than one purchasing scenario.

What should the solutions architect recommend?

- ☐ A Migration Evaluator, importing the existing third-party data, producing TCO with what-if purchasing scenarios
- ☐ B The AWS Pricing Calculator, sized from the inventory
- ☐ C Application Discovery Service, then manual cost modelling in a spreadsheet
- ☐ D Migrate a pilot application and extrapolate its cost

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

Migration Evaluator with imported data
Built for the business case: TCO against current cost, multiple purchasing scenarios, and Windows and SQL licensing analysis. It accepts third-party data.
- B

Pricing Calculator
Prices an AWS design. It does not know the on-premises cost, so it cannot produce the comparison the board asked for.
- C

Discovery plus a spreadsheet
Rebuilds Migration Evaluator by hand, slowly, and a hand-built model carries no external credibility with a board.
- D

Pilot and extrapolate
One application's cost profile rarely generalises across a portfolio, and it delays approval by the length of the pilot.

THE SIGNAL

The comparison is the point. Anything that prices only the AWS side answers half the question a board is actually asking.

A migration is under way using Application Migration Service for servers, Database Migration Service for databases, and a partner tool for a large file estate. The programme manager cannot answer how many of the 340 applications are complete, and builds a spreadsheet by hand each Friday.

Each tool has its own console.

What should the solutions architect recommend?

- ☐ A AWS Migration Hub, grouping servers into applications and aggregating status from all three tools
- ☐ B A CloudWatch dashboard combining metrics from each tool
- ☐ C Continue the weekly spreadsheet, but automate its generation
- ☐ D Standardise on Application Migration Service for everything

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Migration Hub

Exactly its purpose: aggregate status from MGN, DMS and supported partner tools, grouped into applications rather than servers. Free.

C Automate the spreadsheet

Builds the aggregation Migration Hub provides for nothing, and still reports per server rather than per application.

B CloudWatch dashboard

CloudWatch shows metrics, not migration state. It has no concept of an application being migrated.

D One tool for everything

MGN does not migrate databases or large file estates well. The tools are different because the problems are.

THE SIGNAL

Grouping into applications is the part that matters. Per-server progress does not answer whether an application has moved.

Dependency analysis of 340 applications shows one cluster of 14 applications sharing a database and messaging bus, and 60 applications with no dependencies beyond Active Directory. Leadership wants the most complex, highest-value application migrated first to prove the programme's value.

The team has never performed a migration.

What should the solutions architect advise?

- ☐ A Begin with low-dependency applications to prove the process, and keep the 14-application cluster in a single later wave
- ☐ B Begin with the most complex application, as leadership requested
- ☐ C Split the 14-application cluster across several waves to spread the risk
- ☐ D Migrate all 340 applications simultaneously to avoid hybrid running costs

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** Low dependency first, cluster together
Wave one's deliverable is a working process and a trained team. The tightly coupled cluster moves together, in one later wave.
- B** Most complex first
Puts the highest-value application in the hands of a team that has never migrated anything. It proves value by risking the most.
- C** Split the cluster
Spreads risk by guaranteeing it. Fourteen tightly coupled applications split across waves run chatty traffic over the hybrid link for months.
- D** All at once
Avoids hybrid cost by concentrating every risk into one event, with a team that has no experience of the process.

THE SIGNAL

Splitting a tightly coupled cluster is the trap. It sounds like risk management and creates a long hybrid period instead.

A team sets up Migration Hub and selects us-east-1 as the home region during initial configuration, without discussion. The migration will move workloads into eu-west-1 and eu-central-1, and the company has a policy that migration metadata must be stored in the EU. Discovery data has already been collected.

What should the solutions architect advise?

- ☐ A The home region cannot be changed; the account must be reconfigured or a different account used, and discovery data re-collected
- ☐ B Change the home region to eu-west-1 in the Migration Hub console
- ☐ C Migration Hub data is automatically replicated to the migration target Regions
- ☐ D Set a second home region for the EU

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

It cannot be changed
The home region is set once per account and is permanent. Meeting the policy means a different account, and re-collecting discovery data.
- B

Change it in the console
There is no such operation. This is the assumption the scenario is testing, and it is the reason the choice matters on day one.
- C

It replicates
Migration Hub data stays in the home Region regardless of where migrations run. That is precisely the problem here.
- D

A second home region
One home region per account, by definition. There is no second.

THE SIGNAL

A one-off, irreversible configuration choice made casually on day one. Worth knowing before somebody clicks it.

What to carry out of this module

- ◆ **Migration Hub and Application Discovery Service are closed to new customers** and still examinable. Answer with them.
- ◆ **Dependency mapping needs the agent.** Agentless gives inventory and utilisation, not what talks to what.
- ◆ **Migration Evaluator builds the business case**, comparing current cost to projected cost with licensing analysis.
- ◆ **Wave one proves the process.** Keep tightly coupled applications in one wave; never split them to spread risk.

ON THE EXAM

Migration Hub's home region is set once and never changed. It is the irreversible decision people make casually.